

Sentinel: An On-Premises Psychophysiological Triage Node for Continuous Mental Health Monitoring

Biomedical Engineering

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This paper describes the engineering design, security hardening, and empirical validation of a low-cost, on-premises platform for continuous psychophysiological monitoring and discrepancy detection in outpatient mental health care.

1. Introduction

Mental health disorders represent one of the most significant global health burdens of the 21st century. The shortage of mental health professionals is particularly acute in low- and middle-income countries. In India, the psychiatrist-to-population ratio is approximately 0.75 per 100,000 people, compared to the WHO minimum recommendation of 1 per 10,000 [1]. Approximately 60 percent of districts lack any mental health services whatsoever, creating a fundamental monitoring gap. A patient receiving outpatient care may see a psychiatrist for one hour per week, leaving 167 hours of unmonitored time during which acute stress, panic episodes, or suicidal ideation can occur without intervention. The National Mental Health Survey of India reported that 80 to 85 percent of individuals with mental health conditions do not receive any treatment, and the economic burden of mental health conditions costs the global economy an estimated 1 trillion USD annually in lost productivity. Technology-assisted monitoring offers a scalable alternative that extends the reach of the existing clinical workforce without requiring a proportional increase in the number of clinicians.

Beyond improving patient monitoring, an equally important engineering objective is supporting the sustainability of outpatient mental health practice. Clinical psychologists and psychiatrists often manage large caseloads while balancing therapy sessions, documentation, follow-up, and administrative responsibilities. As patient demand continues to outpace workforce growth, manually reviewing every patient's condition between appointments becomes increasingly impractical. Rather than replacing clinical judgment, Sentinel was designed to automate routine monitoring, prioritize patients showing meaningful psychophysiological discrepancies, and present clinicians with actionable information that helps them allocate their limited time more efficiently. By reducing repetitive monitoring tasks and directing attention toward patients requiring intervention, the system aims to improve continuity of care while supporting more scalable outpatient mental health services.

Existing digital mental health platforms predominantly rely on cloud-based architectures that transmit patient data to remote servers for storage and analysis. While these systems offer convenience for well-connected clinics, they introduce latency, connectivity dependency, and data sovereignty concerns that are particularly problematic for low-resource or offline clinical environments [2]. A clinic in a rural area with intermittent internet connectivity cannot rely on cloud-based assessment for time-sensitive triage decisions. Furthermore, many existing platforms separate subjective patient-reported outcomes from objective physiological data, creating a blind spot in the clinical picture. The critical signal may lie not in either channel alone but in the discrepancy between them: a patient who reports feeling fine while their physiology indicates high sympathetic arousal may be at elevated risk for decompensation. This concept of psychophysiological discrepancy has been studied in clinical literature but to date has not been implemented as an automated, on-premises triage system suitable for deployment in resource-constrained settings.

Consumer-grade wearable devices such as the Oura Ring, Ultrahuman Ring Air, and Apple

Watch have demonstrated research-grade accuracy for heart rate and heart rate variability monitoring. Validation studies report a mean absolute error of 0.36 BPM for the Oura Ring during sleep and intraclass correlation coefficients of 0.91 for HRV compared to medical-grade ECG [3]. The Apple Watch has shown similarly strong performance across multiple independent validation studies, with HRV measurement error within 5 percent of research-grade Polar H10 chest straps. These devices are already in widespread consumer use, with an estimated 200 million wearables shipped globally in 2025 alone and an installed base exceeding 1 billion units worldwide. Among mental health outpatients in high-income countries, wearable ownership is estimated at 30 to 40 percent and growing, suggesting that a significant fraction of the target patient population already owns compatible hardware. This creates an opportunity to leverage existing patient-owned devices rather than requiring clinics to procure and distribute dedicated medical-grade wearables, which would add significant cost and logistical complexity. The approach also avoids vendor lock-in: if a particular wearable model is discontinued or a patient prefers a different device, the HAL can support both simultaneously without any changes to the core discrepancy detection engine or clinical dashboard.

The clinical concept underpinning Sentinel was informed by structured consultations with three practicing clinical psychologists prior to engineering development. All three affirmed the relevance of the monitoring gap and the potential utility of an automated system that cross-references subjective patient journal entries with objective wearable biometric data to flag incongruence. The clinical workflow—patient journal input, biometric ingestion from consumer wearables, automated discrepancy flagging, periodic psychologist review of flagged entries, and escalation of critical alerts—was subsequently reviewed and endorsed by one medical doctor and one additional clinical psychologist. These consultations informed the system requirements and architectural decisions but do not constitute a formal clinical trial.

This paper presents Sentinel, an on-premises psychophysiological triage node that ingests biometric data from consumer wearables and subjectively reported mental state from patient journals, then applies a rule-based discrepancy detection engine to identify incongruence between subjective report and objective physiology. The system is designed for deployment on low-cost commodity hardware—a single mini-PC or cloud VM at approximately 15 USD per month—and operates entirely within the clinic network boundary with no patient data leaving the premises. The architecture employs three tiers of inference: a local large language model running on the same hardware, a cloud API fallback for when local inference is unavailable, and a deterministic rule-based assessment that requires no external dependencies whatsoever. This tiered design ensures the system always returns a risk assessment within deterministic time bounds regardless of network availability, addressing the offline requirement identified as a critical gap in existing digital mental health platforms [2]. The paper presents the complete engineering design, reproducible benchmarks with IRIS-standard CSV logbook output, and a discussion of limitations and future work. In addition to improving patient continuity of care, the proposed architecture demonstrates how intelligent triage and clinician-centered workflow design can increase the practical scalability of outpatient mental health services without attempting to replace human clinical decision-making.

2. Related Work and Prior Art

Digital mental health monitoring has been approached from several directions in recent literature. Torous et al. [2] provided a comprehensive review of digital mental health platforms and identified data privacy, lack of interoperability, and the absence of offline functionality as critical gaps in existing systems. Their survey of over 1,500 digital health applications found that fewer than 15 percent offered any form of offline capability and fewer than 5 percent used end-to-end encryption for patient data. These findings underscore the need for architectures designed from the ground up for offline operation and data locality.

Wearable-based mental health assessment has been explored in multiple research contexts. The Mindstrong platform demonstrated that smartphone-derived behavioral markers including typing dynamics, scrolling patterns, and social interaction frequency could predict mood changes in patients with major depressive disorder, achieving 82 percent accuracy in mood state classification. The Beiwe platform by Onnela et al. [4] pioneered both active and passive data collection from smartphones for psychiatric research, including GPS-based mobility tracking, phone call and text message metadata, and survey-based self-reports. Both platforms, however, rely on cloud-based processing and do not provide on-premises deployment options for clinics with connectivity constraints or regulatory data sovereignty requirements.

Discrepancy detection as a clinical concept has been studied using ECG and self-report data in controlled laboratory settings. Research by Liao and colleagues demonstrated that mismatches between self-reported anxiety and physiological arousal as measured by skin conductance response predicted treatment outcomes in cognitive behavioral therapy for anxiety disorders. The phenomenon of affective misattribution—where patients report emotional states that differ from their physiological indicators—is well-documented in clinical psychology literature. However, these studies have relied on laboratory-grade equipment and manual or semi-automated analysis. To date, no prior work has automated this discrepancy detection process using consumer wearables and deployed it on low-cost hardware for continuous outpatient monitoring.

On the security front, several frameworks for protecting health data in low-resource environments have been established. The OpenMRS platform provides a reference architecture for offline-capable health information systems used in over 80 countries, demonstrating that SQLite-based storage and containerized deployment are viable for clinic-scale applications. The DHIS2 platform has shown secure data collection in rural African clinics with thousands of concurrent users. Neither platform, however, provides real-time biometric ingestion from wearable devices or AI-based triage capabilities. We are not aware of any prior system that combines on-premises deployment, consumer wearable integration via a hardware abstraction layer, rule-based discrepancy detection, AI-based triage, and defense-in-depth security hardening in a single open-source package. A comparison of Sentinel against existing platforms across key dimensions is presented in Section 5.

It should also be noted that several commercial wellness platforms such as Calm, Headspace, and Woebot offer consumer-facing digital mental health interventions at scale.

These platforms provide evidence-based therapeutic content but are designed as direct-to-consumer wellness tools rather than clinician-facing monitoring and triage systems. They do not ingest wearable biometric data, do not provide clinician dashboards with aggregated patient views, do not offer on-premises deployment options, and do not perform discrepancy-based triage. Sentinel occupies a distinct category as a clinical support tool for outpatient mental health providers, and its architecture reflects this difference in use case and deployment environment.

The gap in the literature that Sentinel specifically addresses is not the absence of wearable-based physiological monitoring, nor the absence of on-premises health information systems, but rather the absence of these capabilities combined with a clinically motivated discrepancy detection mechanism. Existing systems offer either objective biometric monitoring without subjective patient context (pure physiological tracking) or subjective self-reporting without objective physiological validation (standard patient-reported outcome platforms). The synthesis of both channels through a deterministic rule-based engine that cross-references them in real time, deployable on commodity hardware without cloud dependencies, represents the novel contribution that distinguishes this work from prior art.

3. System Architecture and Methodology

This section describes the engineering architecture and design decisions behind Sentinel. The system is organized into a two-container deployment with a FastAPI backend and React frontend, supported by a hardware abstraction layer for wearable integration, a discrepancy detection engine for cross-referencing subjective and objective data, a three-tier AI pipeline for journal summarization and risk assessment, and a defense-in-depth security model incorporating encryption, network isolation, and tamper-evident audit logging.

The following subsections describe each architectural component in detail. The complete system implementation, including all source code and deployment configuration, is maintained in a version-controlled repository with 45 automated tests that verify each component independently as described in Section 4.

3.1 Deployment Architecture

Sentinel is deployed as a two-container Docker Compose application. The backend container runs a FastAPI application (Python 3.11, 12 API endpoints) using uvicorn as the ASGI server. The frontend container runs a React application built with Vite and TypeScript, served by an Nginx reverse proxy that handles TLS termination, static file serving, and route forwarding. The two containers communicate over an internal Docker bridge network with the internal flag explicitly set to true, meaning the backend has no external network access whatsoever and is only reachable through the Nginx reverse proxy on the frontend container. This zero-trust network topology ensures that even if the backend were compromised through an application-layer vulnerability, an attacker could not pivot to the internet, the local area network, or any other container on the host. The frontend container is the sole ingress point: it terminates incoming TLS connections,

forwards only predefined API routes (`/api/*`) to the backend, and returns 404 for all undefined routes.

The database layer uses SQLite 3 with Write-Ahead Log (WAL) journal mode. SQLite was chosen over PostgreSQL or MySQL for deployment simplicity: the entire database is a single file, no separate database server process is needed, backup consists of copying the database and WAL files, and initialization requires zero configuration. For the target deployment scale of 30 patients with an estimated three journal entries per day each (90 entries per day total) and a maximum of approximately 180,000 entries before archival, SQLite's performance characteristics are well within acceptable bounds, as the storage I/O benchmarks in Section 4.4 confirm.

3.2 Hardware Abstraction Layer

Rather than developing custom wearable hardware, Sentinel implements a hardware abstraction layer (HAL) that accepts biometric data from any device exposing heart rate (BPM) and heart rate variability (HRV) via REST or WebSocket. The system defines a standard biometric payload format: `{ bpm: integer, hrv: integer (RMSSD), timestamp: ISO 8601, source: string }`. The HAL normalizes device-specific data into this canonical format, enabling hot-swappable device support without modifying the core detection engine. Adapters are currently defined for the Oura Ring Gen 3 (via Oura Cloud API v2, reporting BPM and RMSSD HRV from photoplethysmography at 1-minute intervals during sleep and 5-minute intervals during wake), the Ultrahuman Ring Air (via Ultrahuman REST API, reporting similar metrics at comparable intervals), and a mock clinical device that generates synthetic biometric data for automated testing. Adding support for a new wearable device requires implementing a single abstract base class with two methods: `connect()` for authentication and handshaking, and `read()` for retrieving the latest biometric sample in the canonical payload format. The HAL is designed to handle intermittent connectivity from wearable devices, which is common with consumer hardware that synchronizes data periodically rather than streaming continuously. Missing data windows are logged but do not trigger alerts, and the system requires at least three consecutive readings within a configurable time window before updating the biometric classification.

This design decision was reached after a failed attempt at custom PCB fabrication using an nRF52840 microcontroller paired with a MAX30102 photoplethysmography sensor for heart rate and HRV measurement. The prototype board, assembled by JLCPCB, suffered from a D+/D- trace routing error on the USB interface that prevented firmware flashing. The failed fabrication attempt consumed approximately six weeks of development time and approximately 400 USD in prototyping and component costs. The pivot to consumer wearables eliminated hardware certification overhead (FDA 510(k) clearance would have been required for a medical device, and FCC certification for wireless communication), reduced per-patient cost from an estimated 150 USD to zero (patients use their own existing devices), and compressed the deployment timeline from an estimated six months to approximately two weeks.

The production HAL is implemented as a pluggable adapter package (`app/services/ring`) built on a single RingSource base contract with a canonical SensorData payload. Three concrete adapters are implemented: `SimulatedRing`, which generates deterministic

per-user, per-hour biometric streams (calm, balanced, and stressed scenarios) for development and testing; VendorAPIRingSource, an adapter base for vendor SDK/cloud interfaces (Oura, Ultrahuman, and similar); and BLEGATTRingSource, a BLE gateway adapter (bleak) that parses the standard Heart Rate Measurement characteristic (0x2A37), reads battery via the standard battery service (0x180F), and provides configurable characteristic maps and byte parsers for proprietary OEM characteristics. All adapters converge on a single authenticated ingestion endpoint (POST /ring/data), so adding a new device never requires changes to the discrepancy engine, crisis engine, or clinical dashboards. Section 7.2 describes the secured device-binding layer that authenticates physical hardware.

3.3 Discrepancy Detection Engine

The core detection algorithm is a rule-based heuristic classifier that operates on two independent channels: text sentiment and biometric state. The design is intentionally deterministic rather than machine-learning-based to ensure explainability, auditability, and fully reproducible behavior across different deployments. Every discrepancy decision can be traced back to specific keywords and biometric thresholds, which is a requirement for clinical accountability.

The sentiment channel uses two hardcoded keyword sets. The positive trigger set contains 18 terms: great, happy, good, wonderful, amazing, fantastic, energetic, refreshed, joy, love, beautiful, perfect, cured, better, peaceful, content, grateful, and optimistic. The negative trigger set contains 22 terms: anxious, scared, terrified, panic, fear, afraid, hopeless, die, kill, suicide, disappear, worried, cannot, unbearable, drowning, alone, numb, struggling, darkness, terrible, falling apart, and can't. A negation detection mechanism strips any keyword preceded by one of 26 negation prefixes (not, no, never, don't, can't, won't, isn't, aren't, wasn't, weren't, haven't, hasn't, hadn't, doesn't, didn't, without, nothing, nobody, nowhere, neither, nor, none, cannot, won't, wouldn't, shouldn't) within a four-word backward window. This resolves the classic "I am not happy" false-positive problem that plagues naive keyword-based approaches: the negation prefix "not" within four words of "happy" cancels the positive trigger, producing a neutral sentiment classification rather than a positive one.

The biometric channel classifies the patient's physiological state into three categories based on heart rate and HRV thresholds. High stress is defined as $\text{BPM} \geq 110$ AND $\text{HRV} \leq 25$. Low stress is defined as $\text{BPM} \leq 80$ AND $\text{HRV} \geq 55$. All other combinations are classified as moderate. These thresholds are prototype design parameters derived from published norms for resting heart rate (60–100 BPM for adults) and HRV RMSSD distributions (20–70 ms in healthy adults) [5], adjusted to create clear separation between the three classification bands. They have not been clinically validated and would require calibration for specific patient populations based on age, sex, fitness level, and comorbidities in a production deployment.

A discrepancy is flagged according to a hardcoded truth table covering all nine sentiment-by-biometric combinations. Discrepancies are triggered in four cases: positive text sentiment with high-stress biometrics, negative text sentiment with low-stress biometrics, negative text sentiment with moderate biometrics, and neutral text sentiment

with high-stress biometrics. The combination of low stress with neutral text does not trigger a discrepancy, as this represents a physiologically congruent healthy state. The combination of low stress with positive text is also considered congruent. The design prioritizes recall over precision: it is intentionally more permissive in flagging discrepancies, accepting a controlled false positive rate in exchange for minimizing false negatives in a safety-critical context.

The complete truth table governing discrepancy decisions is as follows. Positive text with high-stress biometrics produces a discrepancy, as the patient reports well-being while their physiology indicates stress. Positive text with moderate or low-stress biometrics produces no discrepancy, as subjective and objective channels are congruent. Negative text with high-stress biometrics produces no discrepancy, as the patient's subjective distress matches their physiological state (congruent distress). Negative text with moderate or low-stress biometrics produces a discrepancy, as the patient reports distress without a supporting physiological signal. Neutral text with high-stress biometrics produces a discrepancy, indicating physiological arousal that the patient has not subjectively acknowledged. Neutral text with moderate or low-stress biometrics produces no discrepancy.

The engine was validated against 50 hand-crafted test profiles covering all nine sentiment-by-biometric combinations, including edge cases such as empty text strings, zero values for BPM and HRV, crisis-level language, and negation-heavy sentences.

3.4 Three-Tier Artificial Intelligence Pipeline

Journal summarization and risk assessment follow a three-tier fallback architecture designed to guarantee system responsiveness under all network conditions. Tier 1 attempts inference using a local Ollama instance running Mistral 7B, a 7-billion-parameter language model that runs entirely on consumer hardware with 8 GB of RAM. Mistral 7B was chosen over larger models such as Llama 3 70B because its memory footprint allows it to run on the same mini-PC that hosts the Sentinel application without requiring a GPU. A threading lock enforces a minimum 500-millisecond gap between consecutive Ollama requests to prevent thundering herd overload on the host machine. The Ollama instance receives the patient's journal text with a prompt asking for a structured summary including risk level (low, moderate, high), evidence from the journal text supporting the assessment, and suggested clinical follow-up actions. The prompt explicitly instructs the model to classify any indication of suicidal ideation, self-harm, or imminent danger as high risk regardless of surrounding context, and to provide the specific text that triggered the classification.

If Ollama is unavailable (process not running), returns a malformed response (non-JSON output, missing fields), or exceeds a 120-second timeout, Tier 2 attempts inference via the Groq Cloud API. Groq was chosen over alternatives such as OpenAI or Anthropic for its LPU hardware acceleration, which provides inference speeds under 500 milliseconds per query compared to 2–5 seconds for GPU-based providers. The Groq API call uses the same prompt template as Tier 1 but routes to a hosted Mixtral 8x7B model. API credentials are configured through environment variables, not hardcoded in the application.

If both local and cloud AI are unavailable (for example, during a complete network outage), Tier 3 executes a deterministic rule-based assessment using keyword frequency analysis across five clinical categories. Crisis keywords (suicidal, kill, die, suicide, self-harm, hopeless) are scored at 10 per occurrence. High-risk keywords (panic, terrified, drowning, unbearable) are scored at 7 per occurrence. Moderate keywords (anxious, scared, struggling, alone) are scored at 4 per occurrence. Social withdrawal indicators, sleep disturbance indicators, and activity decline indicators are each scored at 3 per occurrence. A composite score of 8 or higher triggers a crisis alert. This deterministic tier executes in under 50 milliseconds and involves no external dependencies, ensuring the system always returns a meaningful risk assessment regardless of infrastructure conditions.

3.5 Encryption and Key Management

Patient data at rest is encrypted using a custom EncryptedText SQLAlchemy TypeDecorator that transparently applies Fernet encryption (AES-128-CBC with HMAC-SHA256 authentication) on every database write and automatic decryption on every read. The encryption key is derived from a clinician-entered passphrase via PBKDF2-HMAC-SHA256 at 600,000 iterations, producing a 32-byte master key. This master key is further split into independent Fernet and HMAC keys via HKDF-Expand with distinct domain separation labels (“encryption” and “authentication”), ensuring that compromise of one derived key does not affect the security of the other.

This design implements a two-factor authentication model at the data layer: identity verification is handled by bcrypt password hashing with a work factor of 12 (something the clinician knows and enters to log in), while data access requires a separate encryption passphrase (something the clinician also knows but enters in a distinct step during session initialization). The two factors are verified independently: the password authenticates the user to the application, while the passphrase decrypts the database contents. A server compromise that exposes the database file, application memory, and filesystem would still not reveal journal contents without the passphrase, because the encryption key exists only in volatile memory and is never written to disk. The benchmark suite measures the full cryptographic round-trip (key derivation plus encryption plus decryption) at 233.2 milliseconds for the chosen 600,000 iteration count, which falls within the NIST SP 800-132 recommended range of 100 to 300 ms for password-based key derivation [6].

3.6 Security Hardening

A systematic penetration test employing OWASP Top 10 methodology, manual SQL injection testing with crafted payloads, cross-site scripting probes across all user-facing input fields, JWT manipulation attempts, rate limiting bypass tests, and automated scanning with the OWASP Zed Attack Proxy identified 22 findings across four severity levels: 4 critical, 7 high, 6 medium, and 5 low. All 22 findings were remediated prior to submission. The findings included three critical-severity issues: a stored XSS vulnerability in the journal entry field (remediated via DOMPurify sanitization with empty allowlists), the absence of rate limiting on the login endpoint (remediated via in-memory sliding window rate limiter), and a verbose error message disclosing SQLite table schemas (remediated via global exception handlers with sanitized user-facing messages). The threat model assumed an

attacker with network access to the clinic LAN but no access to physical hardware or stored database files, and the test scope covered all 12 API endpoints, the Nginx configuration, the frontend rendering pipeline, the JWT implementation, and the cryptographic interface.

Key mitigations include the following. Session management uses `HttpOnly`, `SameSite=Lax` cookies for JWT storage, preventing XSS-based token exfiltration even if an attacker achieves script injection in the browser. A global in-memory rate limiter enforces a sliding window of 100 requests per minute per IP address, mitigating brute-force login attempts and API enumeration. All frontend text-rendering surfaces implement `DOMPurify` sanitization with empty allowlists for both HTML tags and attributes, providing defense-in-depth against both stored and reflected cross-site scripting. Global FastAPI exception handlers log full stack traces server-side for debugging while returning sanitized, user-safe error messages to clients, preventing information disclosure through error messages. HMAC comparisons throughout the codebase use Python's `hmac.compare_digest`, a constant-time comparison function that prevents timing side-channel attacks on authentication checks. The internal Docker network configuration ensures the backend container has no external internet access, eliminating an entire class of data exfiltration attack vectors.

3.7 Data Integrity and Audit

All state-modifying operations are logged to a SHA-256 hash-chained `AuditLog` table. Each row stores the operation type, timestamp, user identifier, affected resource, and the SHA-256 hash of the previous row's concatenated content fields, forming a cryptographic tamper-evident chain. Modifying or deleting any historical row breaks the chain by invalidating all subsequent hashes, providing forensic detection of unauthorized data alteration. A dedicated health check endpoint recomputes the hash chain from the genesis row and reports whether the chain is intact, enabling automated integrity verification. Database resilience is further enhanced through a `backup_wal()` function that copies both the SQLite database file and its Write-Ahead Log to a timestamped backups directory on every application startup, operating alongside SQLite's native WAL journal mode for crash recovery.

3.8 Frontend Design and User Workflow

The React frontend provides distinct role-based interfaces for patients and clinicians, authenticated through separate login routes with role-specific JWT claims. The patient interface supports daily journal entry via free-text input with a 200 to 500 character length guideline, visualization of biometric trends over configurable time windows (24 hours, 7 days, and 30 days), and a color-coded status indicator showing the current discrepancy state (green for congruent, yellow for moderate discrepancy, red for active discrepancy). The clinician dashboard aggregates all enrolled patients on a single page, highlights patients with active discrepancies sorted by severity, displays historical discrepancy frequency patterns per patient, and provides a crisis management interface with a visible countdown timer for escalation responses with color-coded urgency stages (green at >60 seconds, yellow at 30–60 seconds, red at <30 seconds, and flashing red after timeout). The frontend uses responsive CSS media queries targeting smartphone (320–768 px), tablet

(768–1024 px), and desktop (1024+ px) viewport widths, ensuring accessibility across common clinic devices. State management is handled through React Context with separate typed contexts for authentication, patient data, discrepancy state, and crisis management.

3.9 Testing Methodology

The engineering benchmark suite follows IRIS-standard testing conventions with automated setup and teardown of test fixtures. All tests run against an isolated test SQLite database with no network dependencies: the Tier 1 and Tier 2 AI providers are replaced with mock objects that return configurable responses and latencies. Each test records operation timing in milliseconds, system metadata (Python version, SQLite version, CPU model, available RAM), and a pass-fail status. The suite outputs a CSV logbook with all test results, enabling reproducible cross-deployment comparison. The full suite of 45 tests completes in approximately 3 minutes on a Ryzen 5 5600X system with 16 GB RAM.

3.10 Engineering Contribution Summary

The primary engineering contribution is an offline-capable psychophysiological triage architecture that fuses wearable biometric data with patient-reported journal text through a deterministic discrepancy engine, designed specifically for low-resource clinical deployment. The specific contributions are: (a) a negation-aware rule-based classifier achieving 96 percent accuracy with zero false negatives on a 50-profile validation set, (b) a three-tier AI fallback pipeline that guarantees deterministic response times under 50 milliseconds even without network connectivity, (c) a defense-in-depth security model suitable for handling protected health information on commodity hardware, (d) a hardware abstraction layer that supports multiple consumer wearable devices without device-specific integration, and (e) a reproducible 45-test benchmark infrastructure that outputs IRIS-standard CSV logbooks with per-operation timing for academic reproducibility.

4. Empirical Validation and Benchmarks

The benchmark suite consists of 45 automated tests across five categories: discrepancy detection accuracy (12 tests), crisis concurrency scaling (8 tests), storage I/O performance (12 tests), AI provider latency (8 tests), and cryptographic operations (5 tests). All tests use the automated test harness described in Section 3.9. The suite produces an IRIS-standard CSV logbook with per-run timing, system metadata, and pass-fail status for each test.

4.1 Discrepancy Detection Accuracy

The discrepancy engine was tested against a hand-crafted validation set of 50 profiles spanning all nine sentiment-by-biometric combinations. The test set includes 25 profiles with positive text sentiment, 17 with negative text sentiment, and 8 with neutral text sentiment, crossed against high-stress (15 profiles), moderate (20 profiles), and low-stress (15 profiles) biometric readings. Edge cases include empty text strings, zero BPM and HRV values, high heart rate with low HRV (the high-stress signature), and negation-heavy journal entries such as “I am not happy” and “no felt not good today.” Results are summarized in Table 1.

Metric	Value
Total Profiles	50
True Positives	21
True Negatives	27
False Positives	2
False Negatives	0
Accuracy	96.0%
Precision	91%
Recall (Sensitivity)	100%
Specificity	93%
F1 Score	0.95
Per-Profile Latency	<0.1 ms
Negation Accuracy	100% (8/8)

The zero false negative rate is the critical outcome for a safety-critical application where missing a genuine discrepancy could delay clinical intervention. Every genuine discrepancy in the test set was correctly identified regardless of text sentiment complexity, biometric value range, or the presence of negation patterns. The 50-profile validation set was designed to cover all nine sentiment-by-biometric combinations with at least five examples per combination, plus additional edge case profiles for negation handling and threshold boundary conditions. While limited in absolute size, the test set provides complete coverage of the decision space defined by the engine's hardcoded truth table. The two false positives both arose from biometric readings on the threshold boundary: one profile with BPM of 109 and HRV of 24 (one BPM point below the high-stress threshold), and another with BPM of 111 and HRV of 26 (one HRV point above the high-stress threshold). These boundary cases represent inherent ambiguity in any hard-threshold classification system rather than a design flaw. A single-unit change in either measurement would have altered the classification, highlighting the sensitivity of the prototype thresholds and the need for probabilistic rather than binary classification in a production deployment. The negation-aware preprocessing was validated against eight negation test cases covering all 26 negation prefixes and correctly handled all of them, improving overall accuracy from 92 percent to 96 percent compared to the non-negation-aware baseline.

4.2 Cryptographic Performance

Table 2 presents the PBKDF2 key derivation latency at five iteration counts from 10,000 to 600,000. Each data point represents the mean of three runs on a Ryzen 5 5600X. The chosen value of 600,000 iterations (214.3 ms derive time, 233.2 ms full encrypt-decrypt round-trip) falls within the NIST SP 800-132 recommended range of 100 to 300 ms for password-based key derivation [6].

Iterations	Derive (ms)	Encrypt (ms)	Decrypt (ms)
10,000	3.4	0.6	0.6
50,000	17.4	0.3	0.2
100,000	35.1	0.4	0.3
300,000	102.7	0.5	0.3

600,000	214.3	0.6	0.3
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Encryption and decryption latency remain below 1 ms at all iteration counts, confirming that the PBKDF2 iteration count only affects the one-time key derivation step and does not degrade per-operation read-write performance. This separation of concerns is important for usability: the clinician experiences the 233 ms delay only once per session when entering the passphrase, not on every journal entry submission.

4.3 Crisis Engine Stress Testing

The thread-based crisis countdown was tested at concurrency levels of 1, 5, 10, and 25 simultaneous simulated patients to verify that the singleton state machine does not create a performance bottleneck. Results are shown in Table 3.

Concurrent Patients	Overhead (ms)	Crisis Detected	Escalation Completed	False Triggers
1	101	Yes	Yes	0
5	101	Yes	Yes	0
10	102	Yes	Yes	0
25	101	Yes	Yes	0

The overhead remained constant at approximately 101 ms regardless of concurrency level, confirming $O(1)$ scalability for the singleton crisis state machine under the current architecture. The halting protocol, which models escalation from psychologist notification to helpline referral at a simulated 60 seconds of crisis time, completed in 3,265 ms of real time for 65 simulated seconds, representing a 20x time compression ratio used for test feasibility. The 101 ms overhead includes thread creation, state initialization, and the initial crisis assessment call. This benchmark confirms that the crisis engine is not a performance bottleneck even at 25 times the expected single-patient concurrency. While the singleton design limits the system to tracking one active crisis at a time, the constant-time scaling confirms that threading overhead will not be a limiting factor when the state machine is refactored to per-patient instances for the planned September 2026 clinic pilot. The benchmark also confirmed that no simulated patient experienced a missed crisis escalation due to concurrency conflicts, as the singleton state machine uses a locking mechanism that queues escalation events rather than dropping them.

4.4 Storage I/O Benchmarking

SQLite with WAL journal mode was benchmarked against JSON serialization (both plaintext and encrypted) at storage scales of 10, 50, 100, and 500 patient profiles. Each profile contains approximately 2 KB of journal text, biometric readings with timestamps, discrepancy classification results, and metadata. Read and write latencies were measured as the mean of ten operations at each scale. Results are shown in Table 4.

Profiles	SQLite Read (ms)	SQLite Write (ms)	JSON Write (ms)	Storage (KB)
10	10.2	26.9	6.4	130
50	10.8	52.1	19.7	610
100	11.2	88.4	36.2	1,180
500	11.5	195.2	178.5	5,820

SQLite read latency remained approximately 11 ms across all scales from 10 to 500 profiles, demonstrating that read performance at this scale is I/O-bound by the storage medium rather than database size. Write latency scaled approximately linearly from 26.9 ms at 10 profiles to 195.2 ms at 500 profiles, with the growth driven by transaction journaling overhead and B-tree index maintenance. The 12 KB per-profile storage cost yields an estimated database footprint of approximately 360 KB for 30 patients at 10 entries each, well within the constraints of commodity flash storage. JSON serialization, while faster for writes at small scales (6.4 ms at 10 profiles), does not offer queryability, transactional atomicity, or concurrent access guarantees, making it unsuitable for a multi-user clinical application. Extrapolating from these benchmarks, a 30-patient clinic generating three journal entries per patient per day would accumulate approximately 2,700 entries per month with a total storage footprint of approximately 33 MB per year. This is negligible compared to commodity storage capacity and suggests that data archival or purging procedures would not be required for at least 5 to 10 years of continuous operation. This durability guarantee is important for clinical deployments where regulatory requirements may mandate data retention periods of 5 to 10 years.

4.5 AI Provider Latency

The three-tier AI pipeline was benchmarked with simulated provider responses to validate the fallback timing and correctness. Tier 1 (local Ollama with Mistral 7B) was simulated with a 1,200 ms delay matching measured inference time on a Ryzen 5 5600X with 16 GB RAM. Tier 2 (Groq Cloud API) was simulated with a 600 ms delay. Tier 3 (deterministic rule-based) executed immediately. The fallback was tested by programmatically disabling Tier 1 and Tier 2 in the test configuration.

Tier	Provider	Avg Latency (ms)	Status
1	Ollama (Mistral 7B)	1,200 (simulated)	Configurable
2	Groq Cloud API	600 (simulated)	Configurable
3	Deterministic Rule-Based	3.2	Guaranteed
	Mock (Test Harness)	51.5	Passing

The fallback architecture transitions correctly through all three tiers. When Tier 1 is unavailable, the system attempts Tier 2 within the connection timeout window of 120 seconds. When both Tier 1 and Tier 2 are unavailable, Tier 3 returns a risk assessment in under 5 ms. Of the 45 total benchmarks, 43 pass and 2 fail as expected. The two expected failures correspond to the actual (non-mocked) Ollama and Groq API endpoints, which are not running in the isolated test environment. All mock AI provider tests pass with an average latency of 51.5 ms, confirming that the fallback architecture functions correctly when external services are unavailable. Notably, the deterministic Tier 3 assessment (3.2 ms) is approximately 375 times faster than the simulated Tier 1 (1,200 ms) and approximately 188 times faster than the simulated Tier 2 (600 ms), providing a strong argument for keeping the rule-based fallback in the production system even when AI services are available. The design also enables a useful developer workflow: engineers can disable Tier 1 and Tier 2 during development and testing to force deterministic behavior and obtain reproducible test results without AI nondeterminism, then re-enable both tiers for

production deployment to gain the benefit of contextual language understanding. This also means that during normal operation with Tier 1 or Tier 2 active, the total latency for journal processing is dominated by the AI inference time rather than the discrepancy detection or database operations, which together account for less than 15 ms of the end-to-end pipeline.

5. Discussion and Engineering Limitations

The empirical results demonstrate that a rule-based discrepancy detection engine, combined with deterministic AI fallback and hardware-level network isolation, can produce a consistent triage signal on consumer-grade hardware. The 96 percent accuracy with zero false negatives is particularly relevant for a safety-critical clinical application where missing a genuine discrepancy could delay needed intervention. The benchmark suite confirms that the system meets its primary design goals: deterministic behavior, offline resilience, and reproducible performance metrics across the full range of operating conditions.

However, several limitations must be acknowledged. First, the keyword-based sentiment analysis, even with negation-aware preprocessing operating on 26 negation prefixes within a four-word backward window, cannot detect irony, contextual sarcasm, metaphor, clinical jargon, or culturally specific expressions of distress. A journal entry stating “My GAD-7 score was elevated this week” contains no trigger words and would be classified as neutral, potentially missing a genuine anxiety signal. A patient writing “I feel fantastic, but that scares me more than feeling terrible” would trigger a positive sentiment through the keyword “fantastic” while the underlying clinical state is complex and ambiguous. These are inherent limitations of bag-of-words approaches that contextual language models (Tier 1 and Tier 2 of the AI pipeline) are designed to address when available.

Second, the biometric thresholds are prototype design parameters, not clinically validated cutoffs. While derived from published norms for resting heart rate (60–100 BPM) and HRV RMSSD distributions (20–70 ms) [5], they have not been calibrated against clinical outcomes or adjusted for age, sex, fitness level, or comorbid conditions. A healthy athlete may have a resting heart rate of 45 BPM and HRV of 80 ms, which would be classified as low-stress by the current thresholds. An older adult with reduced HRV might read 20 ms at rest, triggering a false high-stress classification. Production deployment would require patient-specific threshold calibration, possibly using the patient’s own historical baseline as a dynamic reference rather than population-level fixed thresholds.

Third, the singleton crisis state machine can only track one active crisis at a time. While this is acceptable for the current single-patient demonstration scenario, a clinic pilot with 30 or more patients would require a per-patient state machine with independent countdown timers, escalation pathways, and resolution tracking. The thread-based architecture supports this extension—the measured 101 ms overhead is independent of concurrency, as confirmed in Section 4.3—but the state management logic has not yet been refactored from singleton to multi-instance.

Fourth, the cryptographic benchmark of 233.2 ms for the full round-trip represents a single sequential key derivation. In a multi-user scenario with concurrent journal submissions, the PBKDF2 derivation becomes a serial bottleneck because the key is derived once per

process initialization and shared across all operations. Migration from PBKDF2 to a memory-hard function such as Argon2id is planned for the production pilot, following OWASP password storage recommendations and the specific guidance in NIST SP 800-63B.

Fifth, the two expected benchmark failures (Ollama and Groq API endpoints) highlight a dependency on external services for AI-powered journal summarization. While the deterministic Tier 3 fallback ensures the system remains functional during network outages, the absence of local AI inference degrades journal summaries from context-aware natural language generation to extractive 200-character truncation. A clinic deploying without a local Ollama instance would lose semantic understanding capability entirely, though the discrepancy detection engine—which operates independently of the AI pipeline and requires no external services—would continue to function at full accuracy.

Sixth, the clinical validation is limited to pre-engineering consultations. The concept was validated by three practicing clinical psychologists prior to development, and the workflow was reviewed by one medical doctor and one additional clinical psychologist. These consultations informed the system requirements, the discrepancy engine design, and the workflow logic, but they do not substitute for a formal clinical trial with human subjects. The system is currently in the bench prototype phase, validated exclusively through in silico testing with simulated patient profiles. Real-world validation with clinical data from patients undergoing outpatient mental health treatment remains as the next development milestone, pending IRB approval.

Compared to related systems, Sentinel occupies a distinct design niche. Unlike the cloud-dependent Mindstrong and Beiwe platforms, Sentinel operates entirely on-premises with no external data transmission. Unlike clinical-grade ECG-based systems, Sentinel uses consumer wearables that patients already own. Unlike research prototypes that require dedicated hardware and supervised lab visits, Sentinel is designed for continuous outpatient monitoring on a 200 USD mini-PC. The trade-offs are deliberate: reduced biometric accuracy compared to medical-grade equipment (consumer PPG versus clinical ECG), reduced analytical depth compared to human-supervised assessment, but dramatically increased monitoring coverage (continuous versus episodic) and lower deployment cost. For the target use case of resource-constrained outpatient clinics seeking to extend their monitoring coverage between sessions, these trade-offs are acceptable. Table 5 summarizes the comparison across key architectural dimensions.

The clinical implication is that Sentinel can extend the monitoring coverage of a single clinician from one hour per week per patient to around-the-clock passive surveillance, at a hardware cost of approximately 200 USD for a 30-patient caseload. While the system cannot replace clinical judgment, it can prioritize clinician attention toward patients whose subjective and objective channels are incongruent, potentially identifying deteriorating patients earlier than scheduled appointments would allow.

An eighth limitation concerns generalizability to different clinical populations. The system was designed and validated in the context of outpatient adult mental health care, but the underlying architecture does not inherently restrict it to this population. Pediatric patients have different baseline heart rate and HRV norms requiring threshold recalibration. Patients

with cardiovascular conditions such as atrial fibrillation or those taking beta-blockers would have altered heart rate and HRV responses that the current prototype thresholds do not account for. Deployment in inpatient, residential, or telepsychiatry settings would require modifications to escalation pathways and crisis response protocols. Each scenario represents a configuration parameterization rather than an architectural redesign, but validation data for these populations does not yet exist.

Ninth, the system has not undergone formal software certification or medical device regulatory review. Sentinel was developed following general secure coding practices but has not been subjected to IEC 62304 (medical device software) or ISO 13485 (quality management) certification. Deployment in regulated healthcare environments would require additional validation of the development lifecycle, formal verification of cryptographic implementation against side-channel attacks, and documented risk management per ISO 14971.

Tenth, data privacy considerations beyond encryption deserve mention. The on-premises architecture ensures data never leaves the clinic network, but clinician access to patient data within the clinic is governed only by standard authentication. A production deployment would benefit from role-based access logging, automatic session timeout, and integration with existing clinic identity management systems.

Seventh, the 50-profile validation set, while covering all nine sentiment-by-biometric combinations and multiple edge cases, is limited in size and diversity. All profiles were hand-crafted by the engineering team rather than sourced from real patient data, introducing potential author bias in profile construction. The test set does not include non-English journal entries, culturally specific expressions of psychological distress, or entries from patients with comorbid physical health conditions such as cardiovascular disease or chronic pain that affect heart rate and HRV independently of mental state. A production-grade validation corpus would require several hundred annotated entries from diverse demographic and clinical populations, collected during an IRB-approved study.

Dimension	Mindstrong [4]	Beiwe [4]	OpenMRS	Sentinel
Deployment	Cloud	Cloud	On-Prem	On-Prem
Data Locality	External	External	Local	Local
Wearable Support	No	No	No	Yes (HAL)
Offline Capable	No	No	Yes	Yes
AI Summarization	No	No	No	Yes (3-Tier)
Discrepancy Detection	No	No	No	Yes
End-to-End Encryption	Partial	Partial	Partial	Yes
Tamper-Evident Audit	No	No	No	Yes
Cost (Monthly)	Per-Seat	Per-Seat	Free	~\$15
Hardware Required	Smartphone	Smartphone	Server	\$200 Mini-PC

Sentinel is the only system in this comparison that combines on-premises deployment with wearable integration, offline-capable AI summarization, and discrepancy-based triage. Every other platform addresses at most two of these four requirements. The trade-off is that Sentinel operates at a smaller deployment scale (30 patients versus thousands) and uses

consumer-grade biometric sensors rather than medical devices.

From a cost perspective, the total hardware and infrastructure cost for a 30-patient Sentinel deployment is approximately 200 USD for the mini-PC (one-time) plus 15 USD per month for cloud VM hosting if the clinic does not have on-site hardware. The per-patient cost is approximately zero for hardware (patients use their own wearables) and approximately 0.50 USD per month for infrastructure. This compares favorably to cloud-based platforms that charge 10 to 50 USD per patient per month, and to clinical-grade monitoring systems such as Holter monitors that cost 500 to 2,000 USD per device with limited recording durations. The cost advantage is enabled by the decision to use consumer wearables and commodity computing hardware rather than medical-grade equipment.

An important consideration for deployment is the total cost of ownership beyond hardware. The Sentinel system requires no per-patient licensing fees, no cloud subscription for core functionality (AI Tier 3 is built-in and free), and no specialized IT staff for maintenance beyond basic Docker and networking knowledge. The primary ongoing cost is the clinician time required to review flagged discrepancies, which is estimated at approximately 5 to 10 minutes per patient per week based on the workflow consultation. For a 30-patient caseload, this translates to 2.5 to 5 hours of clinician time per week, which is within the typical allocation for between-session monitoring in outpatient mental health care.

6. Conclusion and Future Work

Sentinel demonstrates that an on-premises psychophysiological monitoring platform can be engineered at low cost while incorporating security measures appropriate for protected health information. The system addresses a specific gap in the current digital mental health landscape: the absence of a low-cost, offline-capable, discrepancy-based triage platform that fuses wearable biometrics with patient-reported mental state. The key contributions are: (1) a negation-aware rule-based discrepancy engine achieving 96 percent accuracy with zero false negatives on a 50-profile validation set, (2) a three-tier AI fallback architecture that guarantees system responsiveness regardless of network conditions, (3) a defense-in-depth security model incorporating encrypted at-rest storage, constant-time cryptographic operations, network-level isolation, and input sanitization, (4) a hardware abstraction layer enabling multi-vendor wearable support without device-specific code changes, and (5) a reproducible 45-test benchmark suite with IRIS-standard CSV logbook output for academic verification.

The immediate development roadmap includes four priorities. First, migration from bcrypt to Argon2id for password hashing and replacement of PBKDF2 with Argon2id in the key derivation pipeline, addressing the known limitation of non-memory-hard functions against ASIC- and GPU-based brute-force attacks. Argon2id will also reduce the serial bottleneck identified in Section 5 by allowing configurable parallelism and memory hardness parameters tailored to the deployment hardware.

Second, refactoring the crisis state machine from singleton to per-patient instances with independent countdown timers, escalation pathways, and resolution tracking. The threading overhead has been confirmed as $O(1)$ with respect to concurrency (Section 4.3), so the refactored design is expected to scale linearly to at least 50 concurrent patients on

the target mini-PC hardware. This refactoring is the critical path dependency for the planned September 2026 clinic pilot with 30 patients.

Third, integration of a continuous integration pipeline via GitHub Actions for automated benchmark regression testing on each commit. The existing 45-test suite completes in approximately 3 minutes on commodity hardware, making it suitable as a pre-merge check. The CI pipeline will also generate and archive the IRIS-standard CSV logbook for each run, creating a performance history that can be queried for regression analysis.

Fourth, development of a clinical data collection protocol for IRB-approved validation with human subjects. The protocol will include both healthy controls for baseline physiological data collection and patients undergoing outpatient mental health treatment for discrepancy detection validation. The study design will compare Sentinel's automated discrepancy classifications against clinician assessments for the same journal and biometric data, providing the first real-world accuracy measurement of the system.

Fifth, development of a companion mobile application for patient journal entry and wearable data synchronization, reducing reliance on desktop browser access and improving the patient experience for daily journal submissions.

Long-term, Sentinel is designed for turnkey deployment: a 200 USD mini-PC pre-loaded with the Docker Compose configuration, requiring only LAN connectivity and clinician credential configuration. The hardware abstraction layer ensures compatibility with any consumer wearable that exposes heart rate and HRV data, future-proofing the platform as wearable technology continues to evolve. All source code, benchmark data, engineering logbooks, and technical documentation are maintained in a version-controlled repository to facilitate academic reproducibility and community contribution. The system is released under an open-source license. The open-source model was deliberately chosen, following the principles of reproducible research in computational biomedicine, to enable independent security auditing by third-party researchers, community-driven adapter development for new wearable devices, and collaborative extension of the discrepancy engine to support additional languages and cultural contexts of distress expression.

The total development effort for Sentinel spanned approximately 14 weeks from initial concept to the current benchmark prototype, including the failed PCB fabrication attempt, the clinical validation consultations, the core engineering and security hardening, and the benchmark infrastructure. Of this, approximately six weeks were consumed by the PCB fabrication attempt, which ultimately informed the pivot to consumer wearables. The actual software development and integration required approximately eight weeks, suggesting that the system can be reproduced and deployed by an independent team with comparable resources in approximately two months.

Since the initial benchmark prototype, Sentinel has received a funding award from Emergent Ventures to support hardware procurement and pilot deployment. The hardware path is being executed through an OEM smart ring sourced from Jport (China), which exposes a vendor SDK/API and BLE interface. Post-submission engineering (described in Section 7) added a secured wearable ingestion layer with per-device authentication, a pluggable ring SDK covering simulated, BLE, and vendor-cloud adapters, an installable progressive web application, and a dual-mode AI companion output. These extensions

advance the system toward the planned clinic pilot with 30 subjects.

7. Post-Submission Engineering Extensions

Following the initial benchmark prototype, Sentinel was extended in four areas: a secured wearable ingestion layer, a pluggable ring SDK, a dual-mode AI companion output, and progressive web application deployment. All extensions were validated through the existing test infrastructure and an end-to-end device-to-database ingestion check described below.

7.1 Secured Wearable Ingestion and Device Binding

Physical wearables must not authenticate with a patient's password. Sentinel implements a device-binding layer backed by a `ring_devices` table. A patient pairs a device serial through `POST /ring/pair`, which returns a one-time device token. Only the SHA-256 hash of the token is stored at rest, and tokens are validated with a constant-time comparison (`hmac.compare_digest`). Each sensor push supplies `X-Device-Serial` and `X-Device-Token` headers; the ingestion endpoint resolves the owning patient, updates a last-seen timestamp, and records the reading. A revoked device (`POST /ring/unpair`) is rejected immediately because the authentication query filters on active status, and re-pairing a revoked serial re-issues a fresh token. The patient-JWT path remains available for software clients and simulated pushes. End-to-end verification confirmed correct rejection of wrong tokens, unknown serials, and revoked devices (`HTTP 401`) alongside successful authenticated pushes.

7.2 Pluggable Ring SDK

The ring SDK (`app/services/ring`) defines a `RingSource` base contract and a canonical `SensorData` payload, with three adapters. `SimulatedRing` produces deterministic per-user, per-hour biometric streams across calm, balanced, and stressed scenarios, seeded so that a given patient and hour always reproduce the same readings. `VendorAPIRingSource` wraps vendor SDK/cloud interfaces behind a single `_fetch()` hook. `BLEGATTRingSource` implements BLE gateway ingestion using the `bleak` library, parsing the standard Heart Rate Measurement characteristic (8-bit and 16-bit formats), reading battery level via the standard battery service, and exposing configurable characteristic maps plus per-characteristic byte parsers for OEM-proprietary characteristics. A generic bridge script (`scripts/ring_bridge.py`) runs any adapter on a polling loop and pushes readings through the device-token ingestion path, so the same code base serves simulated demos, BLE rings, and vendor-cloud rings.

7.3 Dual-Mode AI Companion Output

The AI summarization layer now generates two distinct outputs from the same journal entry. For the patient, the model produces a warm, supportive summary in the voice of a friendly AI companion, explicitly instructed to avoid clinical language and unsolicited advice. For the psychologist, the model produces a structured clinical summary (OAP format) suitable for pre-session review. Both outputs derive from the same three-tier pipeline, preserving the deterministic fallback guarantees described in Section 3.4.

7.4 Progressive Web Application

The frontend was converted to an installable progressive web application with a web manifest, installable icons, and a service worker implementing network-first caching for API requests and stale-while-revalidate caching for static assets. The application can be installed to the home screen on mobile devices and provides an offline application shell, reducing reliance on desktop browser access for daily journal submissions.

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AI Utilization Disclosure

This manuscript was edited for syntax refinement, grammar correction, and language formatting using a large language model (Anthropic Claude, accessed via the OpenCode interface). No generative AI was used for the conception of the research question, the design of the engineering architecture, the writing of source code, the analysis of benchmark data, or the formulation of conclusions. The AI tool was used exclusively as a text editor to improve the clarity and linguistic quality of the authors' original technical writing.